

Data Consistency Techniques

Data consistency is a key aspect of data cleaning, directly affecting the reliability and accuracy of your analysis. Common consistency issues include **inconsistent categories**, such as variations in spelling or capitalization within a column that should contain uniform values; **mixed data types**, where a single column contains both strings and numbers, making calculations or grouping unreliable; and **formatting errors**, such as inconsistent date formats or misplaced whitespace. These problems can lead to misleading results or errors in downstream analysis if not properly addressed.

```
import pandas as pd

data = {
    "City": ["New York", "new york", "Los Angeles", "los angeles",
            "Chicago", "CHICAGO"],
    "Population": [8000000, "8000000", 4000000, "4000000", 2700000,
                  "2,700,000"]
}

df = pd.DataFrame(data)
print(df)
```